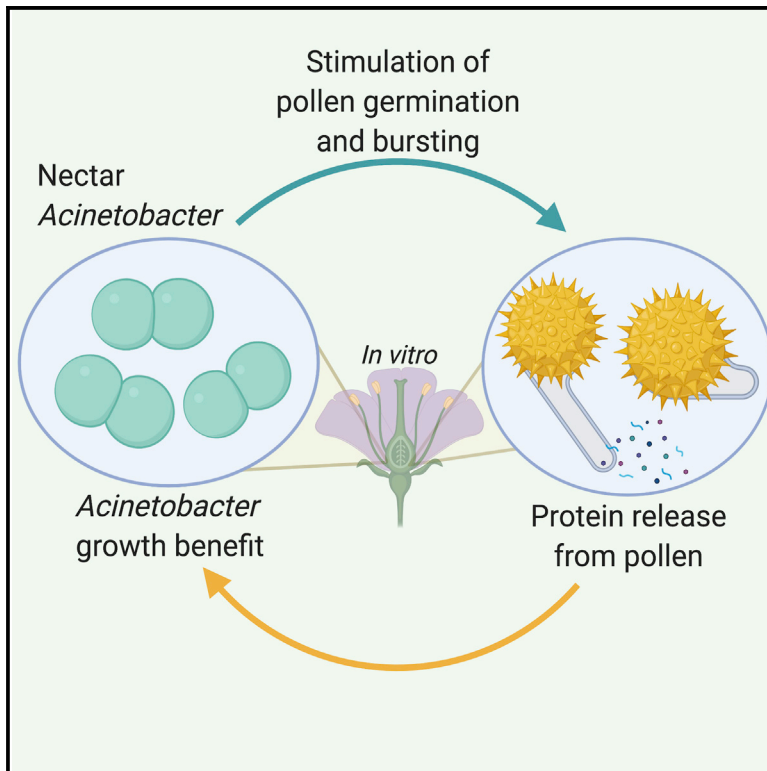


Current Biology

Nectar bacteria stimulate pollen germination and bursting to enhance microbial fitness

Graphical abstract



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In brief

Pollen is nutrient rich but protected by the pollen wall, which hinders pollen digestion. Nectar microbes are nitrogen limited and frequently encounter pollen. Christensen et al. show that nectar-dwelling *Acinetobacter* bacteria stimulate protein release by inducing pollen to germinate and burst, benefitting *Acinetobacter*.

Highlights

- Nectar bacteria *Acinetobacter* induce pollen germination and bursting
- Pollen germination and bursting increase protein in solution
- Germinability of pollen benefits *Acinetobacter*, but not nectar yeast



Report

Nectar bacteria stimulate pollen germination and bursting to enhance microbial fitness

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SUMMARY

Many organisms consume pollen, yet mechanisms of its digestion remain a fundamental enigma in pollination biology,^{1–3} as pollen is protected by a recalcitrant outer shell.^{4–8} Pollen is commonly found in floral nectar,^{9,10} as are nectar microbes, which are nearly ubiquitous among flowers.^{11–13} Nectar specialist bacteria, like *Acinetobacter*, can reach high densities (up to 10⁹ cells/mL), despite the fact that floral nectar is nitrogen poor.^{14–17} Here, we show evidence that the genus *Acinetobacter*, prevalent nectar- and bee-associated bacteria,^{12,18–20} can induce pollen germination and bursting, gain access to protoplasm nutrients, and thereby grow to higher densities. Although induced germination had been suggested as a potential method in macroscopic pollen consumers,^{2,21–23} and fungal inhibition of pollen germination has been shown,^{24–27} direct biological induction of germination has not been empirically documented outside of plants.^{28–32} *Acinetobacter pollinis* SCC477¹⁹ induced over 5× greater pollen germination and 20× greater pollen bursting than that of uninoculated pollen by 45 min. When provided with germinable pollen, *A. pollinis* stimulates protein release and grows to nearly twice the density compared to growth with ungerminable pollen, indicating that stimulation of germination benefits bacterial fitness. In contrast, a common nectar-inhabiting yeast (*Metschnikowia*)³³ neither induced nor benefited from pollen germination. We conclude that *Acinetobacter* both specifically causes and benefits from inducing pollen germination and bursting. Further study of microbe-pollen interactions may inform many aspects of pollination ecology, including floral microbial ecology,^{34,35} pollinator nutrient acquisition from pollen,^{2,3,21,36} and cues of pollen germination for plant reproduction.^{37–39}

RESULTS AND DISCUSSION

Microbial species and strains differ in effects on *E. californica* pollen germination and bursting

Nectar microbes may gain access to pollen nutrients by causing pollen to prematurely germinate.^{40,41} To assess microbial effects on *Eschscholzia californica* pollen germination, we inoculated pollen solutions with seven microbial species (detailed in STAR Methods) and imaged them at 15, 45, and 90 min (n = 27–36 per species). Nectar- and bee-associated *Acinetobacter* species and strains were selected following initial observations with pollen and because they are both common in nectar.^{9–12,18–20} *Metschnikowia reukaufii*, a common nectar yeast, frequently co-occurs with^{9,10} and benefits from pollen³⁵ but is associated with inhibitory effects on pollinia germination.⁴² *Pectobacterium carotovorum* was included to control for bacterial presence (microbial-associated molecular patterns may trigger plant immune response) and to test the potential mechanism of pectinase activity,⁴³ which can stimulate pollen germination and bursting,⁴⁴ and is associated with pollen degradation.^{45–47} In the assays, a pollen grain was classified as germinated if intine was visibly bulging out of the exine, regardless of whether the pollen tube

had also burst (Figures 1C and 1D). The germination percentage for each replicate was calculated by dividing grains that had germinated by total grains and multiplying by 100.

By 15 min, pollen in wells inoculated with *Acinetobacter pollinis* SCC477 or *A. boissieri* had, on average, at least 3× higher germination than the control uninoculated pollen (6.4%, 8%, and <2%, respectively), and germination was significantly higher than pollen inoculated with *P. carotovorum* (<1%) or *M. reukaufii* (<1%) (Kruskal-Wallis $\chi^2 = 46.366$, df = 7, $p < 0.0001$; pairwise comparisons $p < 0.001$; Figures 2A, 2C, and 2D). Across all microbial treatments, pollen germination increased over time, but wells treated with *A. pollinis* SCC477 and *A. boissieri* had, respectively, 4.4× and 3.7× higher average pollen germination than the uninoculated pollen at 90 min (36.9%, 31.3%, and 8.4%, respectively; Kruskal-Wallis $\chi^2 = 53.702$, df = 7, $p < 0.0001$; pairwise comparisons $p < 0.001$; Figures 2A, 2C, and 2D). In contrast, neither the yeast *M. reukaufii* nor pectinase producer *P. carotovorum* affected pollen germination differently from the uninoculated control at any time point ($p > 0.05$). All *Acinetobacter* strains induced significantly higher germination compared to the control by 90 min except for *A. nectaris*, which had variable effects on pollen germination (Figure 2A).



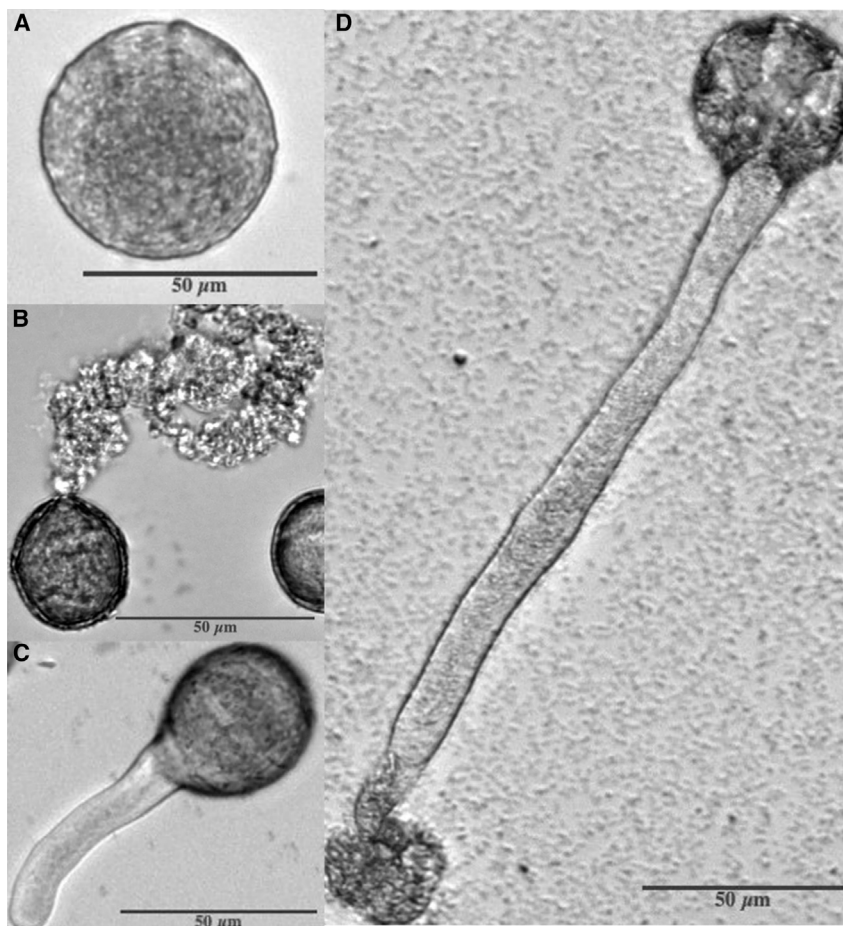


Figure 1. Observed states of *Eschscholzia californica* (California poppy) pollen grains

(A) Whole, ungerminated pollen grain.
(B) Burst pollen grain releasing protoplasm without signs of pollen tube.
(C) Germinated pollen grain.
(D) Pollen grain that germinated, and subsequently the tip of the pollen tube burst, releasing protoplasm.

See also [Video S1](#)

lower bacterial densities (Figures S1A–S1C). Pollen tubes also grew to shorter lengths when inoculated with higher concentrations of SCC477 (Figure S1D).

***A. pollinis* SCC477, but not *M. reukaufii*, gains additional benefit from germinability of pollen**

To assess the effects of pollen presence and germination on microbial growth, we assessed the growth of two species that differed in their ability to germinate pollen: *A. pollinis* SCC477, which readily germinates and bursts pollen, and the yeast *M. reukaufii*, which did not germinate or burst pollen in our assays (Figure 2). We made microbial suspensions that were within the normal range of microbial densities in nectar,^{12,14} then added no pollen, germinable pollen (GP), or pollen rendered unable to germinate (UP, confirmed visu-

To assess microbial induction of pollen bursting, which releases nutrient-dense protoplasm,³ we also monitored pollen tube and whole pollen bursting visually in the same assays referenced above for pollen germination. A pollen grain was classified as burst if it released protoplasm either from the pollen tube (tip bursting) or directly from the pollen grain (Figures 1B and 1D). Both *A. pollinis* SCC477 and *A. boissieri*-inoculated pollen showed significantly higher pollen bursting than the no-microbe control, *M. reukaufii* ($p < 0.001$), or *P. carotovorum* ($p < 0.05$) by 45 min (Kruskal-Wallis $\chi^2 = 47.557$, $df = 7$, $p < 0.0001$; Figure 2B). The average percentages of burst pollen grains for *A. pollinis* SCC477 and *A. boissieri* were 75% and 45% higher than the no-microbe control at 45 min, respectively. At 90 min, pollen inoculated with *A. boissieri*, *A. apis*, *A. pollinis* SCC477, and *A. pollinis* SCC474 were burst to a greater extent than in the no-microbe control (Kruskal-Wallis $\chi^2 = 74.454$, $df = 7$, $p < 0.0001$; pairwise comparisons $p < 0.001$; Figure 2B), but *A. nectaris*, *M. reukaufii*, and *P. carotovorum* did not differ significantly from the no-microbe control at any time point (Figure 2B).

In a separate assay, we used a dilution series of SCC477 to determine whether concentration of SCC477 had an effect of pollen germination and bursting over 24 h. We found that bacterial effects on germination, bursting, and pollen tube length vary with bacterial titer. Pollen inoculated with the highest densities of SCC477 germinated and burst more quickly than those with

ally at 24 h). Compared to microbial growth with no pollen, *A. pollinis* SCC477 and *M. reukaufii* displayed significantly higher cell density after 24 h with either pollen (ANOVA *A. pollinis*, $F_{4,40} = 56.17$, $p < 0.0001$; ANOVA *M. reukaufii*, $F_{4,40} = 21.15$, $p < 0.0001$; pairwise comparisons $p < 0.0001$; Figure 3). However, only *A. pollinis* SCC477 growth differed with pollen germinability: its cell density after 24 h was almost twice as high with GP (1.9× higher than UP, 7.4× higher than no pollen control) than it was with UP (3.8× higher than no pollen control) (ANOVA $F_{4,40} = 56.17$, $df = 4$, $p < 0.0001$; pairwise comparisons $p < 0.0001$; Figure 3A). In the absence of pollen, both microbes barely grew from the starting concentration during the 24-h window (Figure 3), indicating significant reliance on pollen-derived nutrients. *M. reukaufii* grew equally well with both pollen treatments, indicating comparable nutrient content for GP and UP (Figure 3B). Uninoculated pollen had low levels of contamination, likely from the pollen itself (Figure S4).

***A. pollinis* SCC477 effects on pollen are robust in alternative media**

To validate that *Acinetobacter pollinis* SCC477 effects on pollen are not dependent on specific components of BK media used in assays above, and to examine potential mechanisms of microbial growth germination, we performed additional assays in a second media type. We used 30% sucrose media more closely mimicking

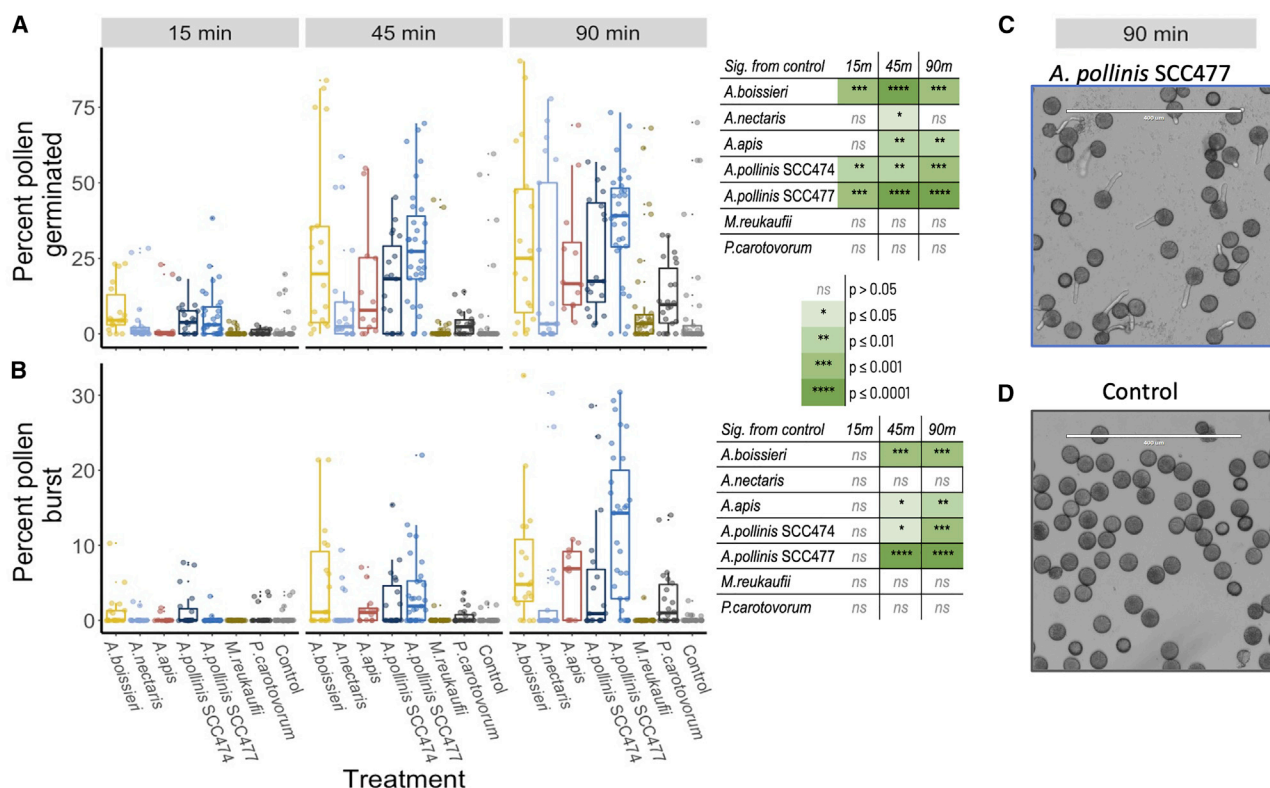


Figure 2. Nectar *Acinetobacter* increase pollen germination and bursting

(A) Percent of pollen grains that germinated with each microbe treatment at 15, 45, and 90 min.

(B) Percent of pollen grains that burst with each microbe treatment at 15, 45, and 90 min.

For (A) and (B), significance from control used Kruskal Wallis test followed by Dunn's multiple comparisons; "ns" indicates no significant difference, $n = 12$ –33, and a total 41,079 pollen grains. Center lines correspond to the median, boxes define interquartile range (IQR), and whiskers extend $+1.5$ IQR. Note difference in scale of y axes.

(C and D) Cropped images of pollen at the 90 min time point when inoculated with *A. pollinis* SCC477 (C) or control with no microbe inoculation (D).

See also Figures S1, S2, and S4, and Video S1.

nectar composition, as BK germination medium is lower in sucrose and contains calcium (Ca) and boron (B), which are typically not found in nectar.^{48,49} To validate microbial effects on pollen, we imaged and quantified pollen germination and bursting at time 0, 12 h, and 24 h. In media lacking Ca and B, SCC477 induced initial stages of pollen germination and pollen bursting by 12 h (Figure S3). Germination was over 10 $\times$ higher with SCC477 inoculated than in the control (35.6% and 3.4% respectively; Figure 4A), and bursting with SCC477 inoculation was nearly 3 $\times$ higher than the control (1.7% and 0.6%, respectively; Figure 4B).

A. pollinis SCC477 stimulates protein release from germinable pollen

In the same assays as described above, we examined the potential cause of the observed growth benefit from GP by quantifying protein levels in solution after exposure of GP and UP to SCC477. Using Modified Lowry Protein Assays, we found that GP exposed to SCC477 released over twice as much protein into solution than uninoculated GP over 24 h, while the addition of SCC477 only marginally increased protein release for UP (Kruskal-Wallis $\chi^2 = 40.768$, $df = 5$, $p < 0.0001$; Figure 4C). Further, GP and UP released near-equivalent amounts of protein when not exposed to SCC477.

Discussion

For many flower-associated organisms, pollen is the primary source of protein and trace nutrients,³ but mechanisms by which consumers breach pollen's recalcitrant outer shell remain poorly documented. Here, we show that the nectar specialist bacteria *Acinetobacter* can induce pollen germination, bursting, and protein release into solution, resulting in higher microbial density when grown with GP. While pollen benefits the growth of most nectar microbes,^{9,10} the results here demonstrate that microbial species differentially induce changes in pollen physiology that affect nutrient release and obtain differential growth benefit from pollen in a habitat that is often nutrient limited.^{15,17,50}

Acinetobacter's ability to access additional pollen nutrients likely contributes to its ecology. The nectar environment is a highly specialized niche, requiring microbial inhabitants to tolerate high osmolarity, plant secondary metabolites or toxins, reactive oxygen species (ROS), and low nutrient levels—all within an ephemeral habitat.^{34,51,52} Here, rapid growth—fueled by pollen nutrients—is likely to strongly affect microbial fitness and allow transfer to the next suitable habitat.^{34,35,53,54} The ability to induce pollen germination and bursting could contribute to the prevalence and abundance of *Acinetobacter* species in floral nectars^{11,12,14} and other environments frequently containing pollen,

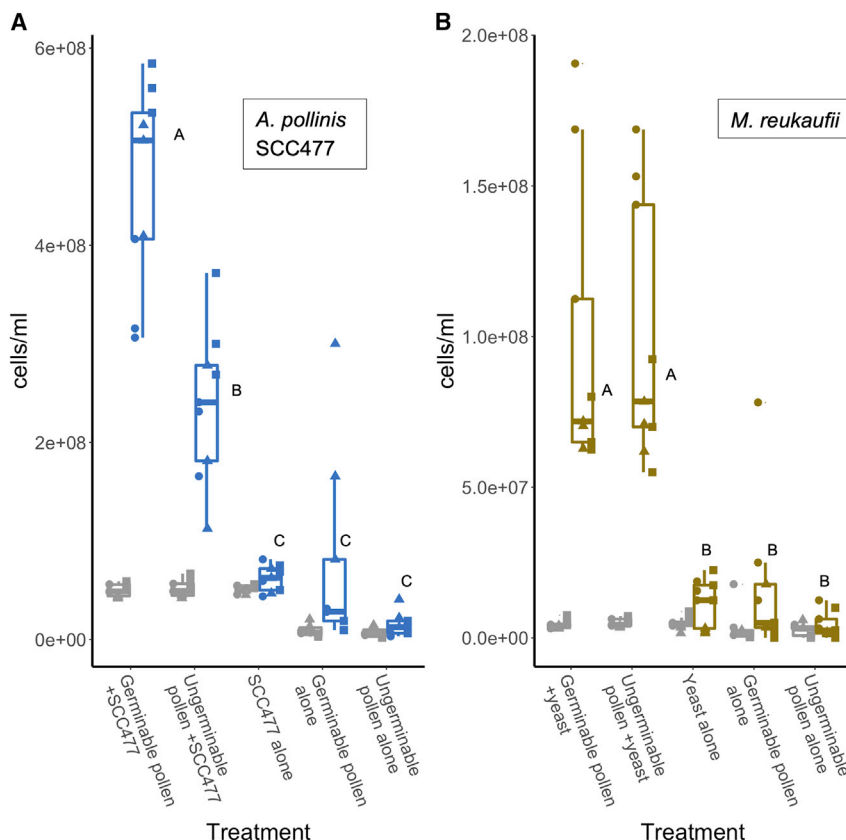


Figure 3. *Acinetobacter* growth is enhanced by germinable pollen addition

Gray indicates time 0, blue and gold indicate 24 h timepoint for *A. pollinis* SCC477 and *M. reukaufii*, respectively. Standard error bars are shown, significance between treatments marked by lettering based on ANOVA followed by Tukey multiple comparisons. Data point shape indicates assay replicate. Center lines correspond to the median, boxes define interquartile range (IQR), and whiskers extend +1.5 IQR. Uninoculated pollen had low levels of microbial growth. Note difference in scale of y axes.

(A) *A. pollinis* SCC477 cell density at time 0 and 24 h. Data represent three assays, three biological replicates of each treatment per assay, $n = 9$.

(B) *M. reukaufii* cell density at time 0 and 24 h. Data represent three assays, three biological replicates of each treatment per assay, $n = 9$.

See also to Figure S4.

such as bee pollen provisions and potentially stigmas.^{55–58} The fact that UP still releases protein and benefits growth could be due to leaching of internal nutrients or from the pollenkit on the pollen exterior. The difference between microbial species' effects on pollen may allow for niche partitioning among nutrient sources. Indeed, the common nectar yeast *M. reukaufii* neither induced germination nor derived additional benefit from germination, suggesting that *Acinetobacter* may have a key fitness advantage in the floral landscape. Alternatively, bacterial effects may facilitate yeast growth.^{50,59} Though we predict that the observed benefit from pollen is due to the observed increased availability of protein (Figure 4), pollen germination or bursting could also release other components including lipids, sterols, trace elements³ or growth hormones.^{60,61} Additionally, the UP may have been altered in other ways that we were unable to characterize, which may also affect microbial growth. The mechanism by which pollen benefits nectar microbes merits further study.

Although the mechanism behind the induced germination phenotype described here has yet to be determined, microbe-pollen interactions could prove a useful system in identifying chemical cues required for pollen germination. Pollen germination is tightly controlled, being influenced by hydration rate of the desiccated grain,⁶² pectin modifications,⁶³ pistil-pollen crosstalk,⁶⁴ calcium,⁴⁹ and other molecules,⁶⁵ as well as aperture number, exine thickness, and nutrient composition.^{39,66} Pollen tube growth requires continuous spatial and temporal adjustments to create and maintain polarity for directed growth; interference can result in tube defects such as branching and premature bursting,^{67–70} similar to the

phenotype observed here (Figure 1). It is worth noting that pollen tube tips have mechanisms for tip bursting to allow bursting sperm release when induced by the female gametophyte during regular fertilization,⁷¹ but are also known to burst *in vitro* due to osmotic changes, polysaccharides (modification, cycling), pH, ROS signaling, and pectinase activity.^{72,73} While we did not identify the mechanism respon-

sible for tube bursting documented here, the published genomes of nectar *Acinetobacter* contain multiple assigned pectinase genes (Table S1), which are known to impact germination and bursting.⁷³ Evidence presented here—including minimal effects of *Pectobacterium*, a known pectinase producer, and a lack of detected pectinase activity by *Acinetobacter*—suggest that pectinases alone are not responsible for the germination or bursting phenotype (Figure S2; Table S1). It is generally observed that pollen germination *in vitro* takes longer than stigmatic germination, presumably due to missing factors or less-than-ideal conditions.⁶⁴ *Acinetobacter* effects may be a promising system to either identify species-specific cues if effects are restricted to *E. californica* pollen or universal cues of pollen germination if *Acinetobacter* can also affect pollen of other plant species. Moreover, the variability within and among strains of *Acinetobacter* (Figure 2) could reflect species specificity in interaction or reliance on pollen,⁵⁰ recent evolution of the phenotype, or variability in pollen state (e.g., desiccation, outdoor temperature, low-abundance microbes, or endophytes that could affect pollen viability). We anticipate that further work on this system will shed light on the specificity and mechanisms involved in interactions between pollen and *Acinetobacter*, as well the potential for other flower-dwelling microbes to alter pollen physiology to access nutrients.

There are ecological implications of microbial effects on pollen that remain to be explored. Since many pollen consumers also consume floral nectar, *Acinetobacter*-induced pollen germination and bursting could illuminate a previously undocumented mechanism for macroscopic species to access pollen nutrients

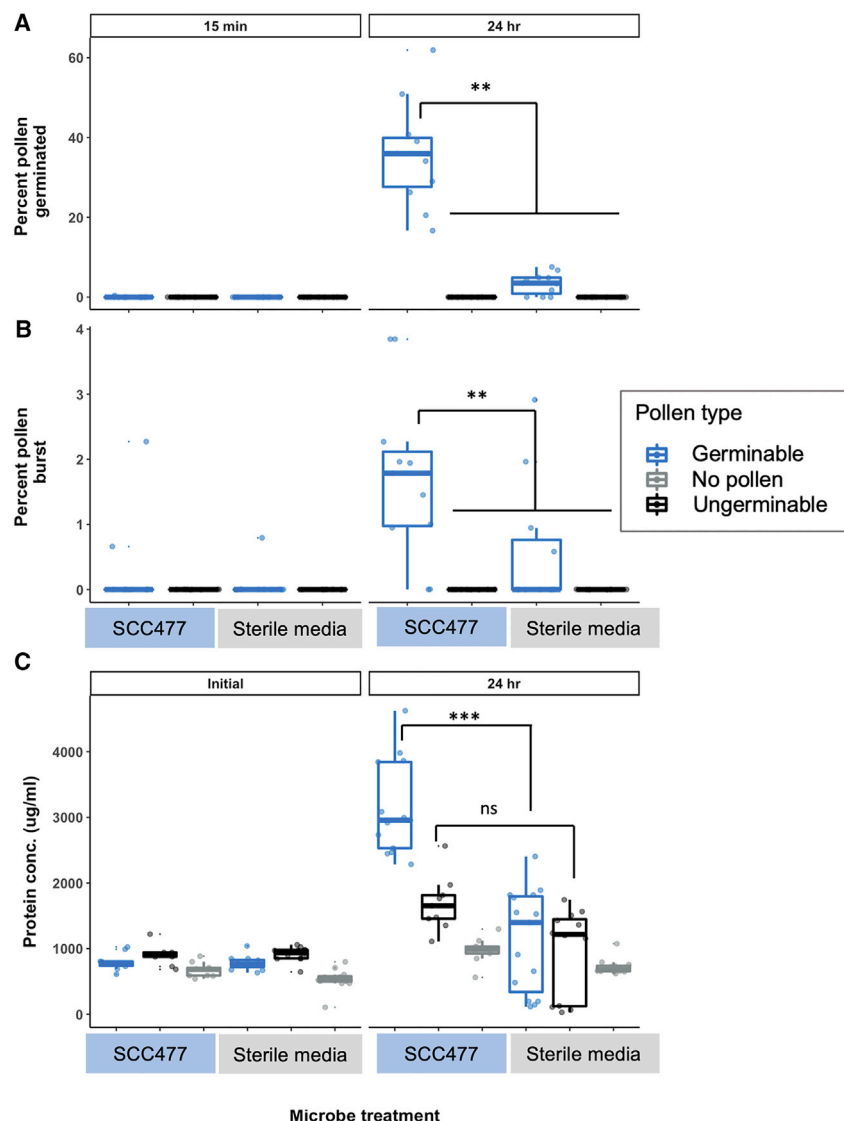


Figure 4. Pollen germination and bursting due to *A. pollinis* SCC477 exposure result in increased release of protein from germinable pollen

Standard error bars are shown, significance between treatments based on Kruskal-Wallis followed by Dunn's multiple comparisons and marked by brackets; "ns" indicates no significant difference ($p = 0.09$). Center lines correspond to the median, boxes define interquartile range (IQR), and whiskers extend ± 1.5 IQR. Data represent three assays, three to eight biological replicates of each treatment per assay, $n = 9$ – 15 , and a total 14,544 pollen grains. Assays done in artificial nectar with 30% (w/v) sucrose. Note difference in scale of y axes.

(A) Pollen germination at 15 min and 24 h in response to *A. pollinis* SCC477.

(B) Pollen bursting at 15 min and 24 h in response to *A. pollinis* SCC477.

(C) Protein concentration in solution at time 0 and 24 h among pollen types with exposure to SCC477.

See also Figures S3 and S4.

via microbial partners. While pollen germination has been considered and studied as a potential mechanism of digestion for macroscopic consumers,^{2,22,23} microbial involvement in this mechanism has not been studied. *Acinetobacter* spp. are common in floral nectar and are also found in the nectar and pollen provisions that bees provide their offspring.^{57,58,74} Our study suggests that the role of floral microbes in pollen germination and bee nutrition warrants further study.

Microbial stimulation of pollen germination may also affect plant fitness. Previous work has shown that nectar microbes can alter pollinator preference for flowers by altering nectar composition and volatiles,^{10,75} directly impacting plant reproduction as these microbes grow to greater density by using pollen nutrients. Effects on pollen nutrient availability to pollinators could also potentially influence pollinator foraging and pollination. Direct effects of microbes on plant reproduction are also possible. *Acinetobacter* is common on many floral surfaces, including stigmas,⁷⁶ and its presence in nectar is correlated with presence in seed microbiomes.⁵⁵ Whether *Acinetobacter* growth on stigmas or seeds

affects pollen or seed germination or success in the context of fertilization will also require further study.

As we have demonstrated, floral microbes can significantly affect pollen physiology—this in turn may have important consequences for plants and pollinators.^{10,34,75,77} Further investigation of microbe-pollen interactions could shed light on the specific biochemical triggers of pollen germination and the ecological effects of microbe-pollen interactions on plants and pollinators.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cub.2021.07.016>.

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AUTHOR CONTRIBUTIONS

S.M.C. and R.L.V. conceived of the study. S.M.C. performed investigation, formal analysis, and visualization, and wrote the original draft. I.M. performed PGA assays and contributed to review and editing. R.L.V. contributed to review and editing, supervision, and final approval for publication.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

One or more of the authors of this paper self-identifies as a member of the LGBTQ+ community. One or more of the authors of this paper received support from a program designed to increase minority representation in science.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
<i>Acinetobacter pollinis</i> SCC477	Isolated from floral nectar ¹⁹	NCBI:txid2605270 ATCC: TSD-214
<i>Acinetobacter pollinis</i> SCC474	Isolated from floral nectar ¹⁹	NCBI:txid2605270
<i>Acinetobacter boissieri</i> ANC4422	Isolated from floral nectar ¹⁸	NCBI:txid1219383
<i>Acinetobacter nectaris</i> BB362	Isolated from floral nectar ¹⁸	NCBI:txid1219382
<i>Acinetobacter apis</i>	Isolated from <i>Apis mellifera</i> , provided by Sergio Álvarez-Pérez ²⁰	NCBI:txid1229165
<i>Metschnikowia reukaufii</i>	Isolated from floral nectar ¹¹	NCBI:txid27327
<i>Pectobacterium carotovorum</i>	Provided by Robert Gilbertson ⁴³	NCBI:txid555
Biological samples		
<i>Eschscholzia californica</i> pollen	UC Davis Arboretum	N/A
Chemicals, peptides, and recombinant proteins		
cetyltrimethylammonium bromide (CTAB)	Chem-Impex	Cat. no 01781
alpha-cyano-4-hydroxycinnamic acid (HCCA)	Sigma Aldrich	Cat. no 70990
Reasoner's 2A agar (R2A)	Thermo Fisher Scientific	Cat. no DF1826171
Yeast extract Media (YM)	Teknova	Cat. no Y0750
Polygalacturonic acid (PGA)	Thermo Fisher Scientific	Cat. no AAL0420614
Tryptone	US Biotech Sources	Cat. no T01PD-500
Soytone	US Biotech Sources	Cat. no SP1PD-500
NaCl	Fisher Chemical	Cat. no 5271-500
D-Fructose	VWR Chemical	Cat. no 0226-1KG
SuperPure Agar	US Biotech Sources	Cat. no A02MG-500
Cycloheximide	ACROS	Cat. no 357420250
Chloramphenicol	Fisher Scientific	Cat. no BP904-100
Phosphate buffered saline (PBS)	Sigma LifeScience	Cat. no D8537-500ML
Sucrose	Fisher Chemical	Cat. no 5271-500
Boric acid	Fisher BioReagents	Cat. no BP168-1
Calcium nitrate	Fisher Scientific	Cat. no AC423530250
Magnesium sulfate	Alfa Aesar	Cat. no 11596
Potassium nitrate	Fisher Scientific	Cat. no AA1344322
Formic acid	Sigma Aldrich	Cat. no F0507-1L
Yeast extract	UC Biotech Sources	Cat. no Y1003-1001
Critical commercial assays		
Pierce Modified Lowry Assay Kit	Thermo Fisher Scientific	Product 23240
Deposited data		
Supplemental data: Nectar bacteria stimulate pollen germination and bursting to enhance their fitness	Dryad Data Repository	Dryad: https://doi.org/10.25338/B8BS75
Supplementary code: Nectar bacteria stimulate pollen germination and bursting to enhance their fitness	Zenodo	Zenodo: https://doi.org/10.5281/zenodo.5123394
Video S1: Nectar bacteria stimulate pollen germination and bursting to enhance their fitness	Zenodo	Zenodo: https://doi.org/10.5281/zenodo.5138284

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
ImageJ/ FIJI2 (v2.0.0)	Schindelin et al. ⁷⁸	https://imagej.nih.gov/ij
Bruker- Ultraflex III FlexControl/ FlexAnalysis (v. 3.4)	Bruker Daltonic	N/A
MALDI Biotyper Compass Explorer (v. 4.1.70)	Bruker Daltonic	N/A
RStudio 1.2.1335	RStudio Team ⁷⁹	https://www.rstudio.com
R 3.6.2	R Core Team ⁸⁰	https://www.r-project.org

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Shawn Christensen (smchristensen@ucdavis.edu).

Materials availability

This study did not generate new unique reagents or materials.

Data and code availability

- All data have been deposited at Dryad Digital Repository and are publicly available as of the date of publication. DOIs are listed in the key resources table.
- All original code has been deposited at Zenodo and is publicly available as of the date of publication. DOIs are listed in the key resources table.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

We chose to examine the nectar *Acinetobacter* clade and their impact on pollen after initial observations of *Acinetobacter* induced pollen germination. We chose 3 nectar-specialist *Acinetobacter* species representing the species most commonly isolated from nectar (*A. boissieri* and *A. nectaris*¹⁸), and an additional species in the clade that was isolated from the honeybee gut (*A. apis*²⁰). We recently isolated and described *A. pollinis*,¹⁹ of which we used two strains. *Acinetobacter pollinis* SCC477 is available through American Type Culture Collection (ATCC), type strain deposit number TSD-214. *Metschnikowia reukaufii* is a commonly isolated nectar-dwelling yeast and is a member of an assemblage of nectar microbes in *Asclepias* that has been observed to negatively impact germination of *Asclepias* pollinia and has also been observed to aggregate around the pollen of *Helleborus fetidus*.^{9,33,42} We included the pectin degrader and plant pathogen *Pectobacterium carotovorum* as a control for Microbial Associated Molecular Patterns (MAMPs: which can trigger plant immune system); it also has high pectinase activity and is a plant associated bacteria that is not associated with nectar or pollen.⁴³

Plants

We used pollen from the California native poppy *Eschscholzia californica* (Papaveraceae). Flowers were obtained from the native plant prairie at the University of California Davis Arboretum, where they receive occasional watering. Flowers were collected in the morning, before 10:00AM, 2-3 days prior to use in assays, which occurred from 2019 to 2021. *E. californica* produces a large amount of pollen, is easily obtainable near our laboratory to allow for minimal effects of storage, and flowers are covered by a cap that is pushed off just before its first opening with the sun; this allowed us to collect flowers before insect or wind activity introduces contaminating microbes (Figure S4). *E. californica* does not have nectar, reducing the likelihood of contaminating nectar microbes. Pollen of Papaveraceae has ectoapertures^{81,82} meaning that the outermost layer of the exine shell is missing from the aperture region but there are still other sporopollenin layers surrounding the intine.⁸³

METHOD DETAILS

Microbial impact on pollen

To test whether microbial inoculation impacts pollen germination and bursting, we collected newly opened *Eschscholzia californica* flowers the morning of the assay to minimize the chances that flowers had been visited by insects or otherwise “contaminated” with

microorganisms or other pollen grains. Many attempts were made to sterilize the pollen prior to the assays (low heat sterilization, autoclaving, chlorine gas fumigation, 50%–70% (v/v) ethanol submersion, 1%–10% (v/v) bleach submersion), but all rendered the pollen unable to form visible pollen tubes at all or were ineffective at sterilizing pollen, so newly collected pollen from pre-collected flowers was used. Newly collected pollen had low levels of microbial growth (Figure 4).

For all microbial growth assays, bacterial strains were plated from freezer stocks onto Tryptic Soy Agar (TSA; 1.5% tryptone, 0.5% soytone, 0.5% NaCl, 5% fructose, 1.5% agar – all (w/v), + cycloheximide) and yeast (*Metschnikowia reukaufii*) onto Yeast extract Media (YM; + chloramphenicol), then colonies were inoculated into Tryptic Soy Broth (same as TSA but -cycloheximide, - agar) and incubated at 25°C for 1–3 days. Suspensions were normalized to an $OD_{600\text{ nm}} = 0.1$ or 0.05 (depending on the microbe with the lowest OD, and consistent within each assay). Suspensions were centrifuged for 5 min at 12,000 rpm to pellet the cells, and supernatant was poured off to remove the growth media along with any released metabolites and proteins, then cells resuspended in phosphate buffered saline (PBS). After resuspension by repeated pipetting, the suspension was vortexed for 30 s and then sonicated for 3 min. To ensure no carryover of metabolites from microbial growth media, these washing steps (from centrifugation through sonication) were performed twice for the first five assays, and three times for the last four assays. The final resuspension was in Brewbaker and Kwack (BK) pollen germination media⁴⁹ (10% w/v sucrose, 100 mg/L boric acid, 300 mg/L calcium nitrate, 200 mg/L magnesium sulfate, 100 mg/L potassium nitrate) instead of PBS. 50 μ L of each washed microbial BK suspension (or the control sterile BK) was added to respective wells (3–5 replicate wells per microbe per assay) in a sterile, flat bottom, clear 96 well plate.

Immediately before beginning each assay, the pollen solution was prepared. Briefly, pollen from 1–3 flowers was vacuumed or shaken directly from anthers into a sterile 5 mL Eppendorf tube and 2 mL of BK solution was added. Immediately after pollen grains were in solution, 50 μ L of the pollen solution was added to each well of the sterile 96-well plate, mixing pollen stock solution every 5–7 wells to ensure even pollen deposition in each well. Control (uninoculated) wells also received 50 μ L of pollen, but instead of 50 μ L of microbe solution, they received 50 μ L of sterile BK. The plate was covered with a lid and wrapped in a layer of aluminum foil to limit evaporation, contamination, and light exposure. Each treatment was replicated in at least three wells per assay. Each well of the plate was then imaged on an EVOS M5000 imaging system at 15, 45, and 90 min after mixing. Final pollen concentrations were on average 20 grains/ μ L in each well, SD = 11 (based on pollen grains/image and extrapolating to volume of well), well within normal range of pollen in nectar.⁹

Image analysis

Images were analyzed in FIJI2 (v2.0.0),⁷⁸ using a custom Macro script (below) to count total pollen grains in each image. The numbers of germinated, burst, and tip burst grains were then counted by hand in FIJI2 using CellCounter.⁸⁴ This was used to count pollen grains within each of three classifications (Figure 1). Any pollen grain touching the edge of the image was excluded in both the total and subset counts. A grain was counted as “germinated” if the pollen intine was clearly emerging from the exine (Figure 1C), as “burst” if the pollen grain was surrounded in loose cytoplasm, but without any visible protrusions (Figure 1B; Video S1), and as “tip burst” if the pollen grain had germinated, and then the pollen tube had burst and released cytoplasm (Figure 1D; Video S1). For all assays, pollen grains were classified twice; once just after assay completion, and all assays were recounted by the same observer within a week prior to combined final analysis to ensure consistency of each category between assays. These final counts were used for analysis. Eleven total assays were done, and two were eliminated from further analysis. One assay was eliminated due to data loss of images-making recount and validation of pollen ratios impossible, and the other was eliminated due to contamination of the initial growth media. An average of 4,564 (+/– 1501.6) pollen grains were counted per assay, with a total of 41,079 pollen grains overall (for microbial comparison assays).

Macro script for counting total pollen grains

We validated this macro script to be accurate with a paired t test comparing macro script counts with hand counts of the same images (90 min time point from three assays). The macro counts and hand counts were not significantly different ($N = 53$, $p = 0.3353$). The 90 min time point was selected because the pollen tubes and burst grains present at 90 min (in some images) were initially an obstacle for accurate macro counting. This script was used for Figure 2 and Figure S1 to count total pollen grains.

All images for each time point were analyzed in a batch using “Process”-> “Batch”-> “Macro” with the following script:

```
run("8-bit");
setAutoThreshold("Default dark no-reset");
//run("Threshold...");
//setThreshold(176, 255);
run("Make Binary," "thresholded remaining black");
run("Convert to Mask");
run("Fill Holes");
run("Watershed");
run("Analyze Particles...", "size=900-3500 circularity=0.30-1.00 show=Overlay display exclude summarize");
```

Note: The size and circularity ranges can be changed for use with different pollen sizes and shapes. This is especially important in the later time points (with germination) as the circularity measure can be altered by protruding pollen tubes, or even bulging pollen tube tips.

Germination data processing

For analysis, the pollen grain counts in each category (described above in main assay description) were converted into percentages of the total number of pollen grains in each image. “Germinated” and “tip burst” were combined in the category of “all germination,” because the “tip burst” pollen had also germinated. Likewise, “burst” and “tip burst” were combined to form the category “all burst” because the pollen grains in both initial categories expelled their protoplasm into the solution. Dataset and code are available in the key resources section.

Density-dependence

To test whether pollen responds differently to differing amounts of *Acinetobacter*, we repeated the above methods, but inoculated pollen with dilution series of *A. pollinis* SCC477 suspensions (1, 1:2, 1:3, 1:5, 1:20, 1:100) as well as sterile BK (no microbes). We started with a solution at an $OD_{600\text{ nm}} = 0.1$ (10^7 cells/mL) and diluted to the above ratios, with the first being “full concentration” and the last consisting of only sterile BK. We then imaged the wells at 15, 45, 90 min, and 3, 4, 6, 12, and 24 h. The images were analyzed, and pollen grains counted and categorized as described for the previous assay. In addition, we measured and recorded the lengths of each pollen tube in all of the images using FIJI2. We measured the tubes using the basic FIJI tool, used the image scale bar to set the scale, then used ROI manager (“Analyze”→“Tools”→“ROI Manager”) with Freehand drawing tool to trace and keep track of pollen tubes (check “Show all”) “add [t]” each line to the manager, then highlight all and “Measure.”

Dataset and code are available in the key resources section.

Microbial growth benefit

To test whether *Acinetobacter* benefits from pollen addition and specifically from pollen germination, we monitored growth responses of *A. pollinis* SCC477 in the presence of live pollen, pollen rendered unable to germinate, and without pollen. We created washed suspensions of $OD_{600\text{ nm}} = 0.1$ *A. pollinis* SCC477 as above (pellet cells, remove media, replace with BK, vortex, sonicate-repeated 3x). *A. pollinis* SCC477 was selected because its addition resulted in the greatest induction of germination and bursting of the species and strains that we tested. The initial cell number of *A. pollinis* SCC477 cell suspension was immediately counted on a hemocytometer and adjusted to 5×10^7 cells/mL.

Pollen was collected as above and was weighed out into each of two 5mL tubes. The first was microwaved for 3 min to prevent germination by submerging the pollen in BK media, opening the lid and placing in a glass beaker to hold tube upright, then immediately microwaving on high for 3 min. The second tube remained untreated. Just before combining at time 0, BK media was added to the fresh, germinable tube of pollen. Assays had consistent concentrations of pollen solution (4.2 mg/mL, ~40 grains/ μ L) which was made by weighing out pollen to nearest hundred-thousandth of a gram and adding enough media to adjust concentration to exactly 4.2 mg/mL.

In 1.7 mL Eppendorf tubes, 50 μ L of *A. pollinis* SCC477 solution was mixed with 50 μ L of the microwaved pollen solution, the live (untreated) pollen, or the no-pollen (BK) control, and 400 μ L of sterile BK ($N = 3$ for each pollen type, final pollen concentration ~4 grains/ μ L). We added the same amount (50 μ L) sterile BK instead of *A. pollinis* SCC477 to 3 tubes each of both untreated and ungerminable (microwaved) pollen as a control for the microbes already present on the pollen. The initial concentration of microbes in each tube was determined by hemocytometer cell count. The tubes were sealed with headspace and left at room temperature for 24 h. At 24 h each tube was sonicated to separate cells for visual enumeration, as nectar-dwelling *Acinetobacter* can form dense biofilms, then vortexed, diluted 1:10 in PBS, and counted on the hemocytometer.

The pollen grains in the tubes that pollen was added to were checked to ensure that the microwaved pollen did not germinate, and that the fresh pollen could. 100 μ L of each tube was also plated on TSA at 24 h to verify that bacterial colony morphology and characteristics matched the inoculum. Individual colonies from these plates were correctly identified by MALDI-TOF and Compass Explorer as described below.

To assess if pollen germinability affects the growth of microbes that do not induce germination, we repeated this assay with *M. reukaufii*, a common nectar yeast. Instead of beginning with 5×10^7 cells/mL, we began with 5×10^6 cells/mL, to control for microbial biomass. Dataset and code are available in the key resources section.

Confirmation of phenotype in alternate media

To confirm that the phenotype of SCC477 induced germination and bursting can occur in ecologically relevant conditions, we used essentially the same setup format as the germination assay (Figure 2), but in artificial nectar media (30% sucrose, 0.125% tryptone, 0.025% soytone- all w/v). This media was selected because it is more ecologically relevant (no boron or calcium, higher sugar content) and allows *Acinetobacter* to grow without the presence of pollen so that we could include controls for just *Acinetobacter* growth and potential resulting protein release. SCC477 was grown for 24 h in artificial nectar then $OD_{600\text{ nm}}$ was measured ($OD_{600\text{ nm}} = 0.1$ to 0.2). The microbes were washed in PBS once (centrifuged at 14,000 rpm for 5 min, decanted supernatant, resuspended in PBS, repeated once with final resuspension in sterile artificial nectar). Pollen solutions for germinable and ungerminable pollen treatments were created as described for growth assays (above) but in artificial nectar instead of BK media (4.2 mg/mL). 50 μ L of microbe solution and 50 μ L of pollen solution were added to each well of a 96 well plate with treatments of germinable pollen, ungerminable pollen or no pollen (sterile artificial nectar; final pollen concentration ~20 grains/ μ L) with SCC477 or without microbial inoculation (sterile artificial nectar).

Pollen germination and bursting were quantified at 15 min, 12 h, and 24 h using the same criteria as germination assay above (germination: visibly protruding intine, bursting: visible release of protoplasm) but on a different microscope due to unavailability of the EVOS M5000 (ECLIPSE Ts2 with Levenhuk M300 Base series microscope camera and Levenhuk Lite software for macOS v 1.016). The resulting images were not compatible for counting with the FIJI Macro script (Supplementary methods 1), therefore we used an iOS app “CountThings”⁸⁵ (Template ID: 386 v. 004) to count total pollen in each image, and validated that the counts were not significantly different from hand counting. Dataset and code are available in the key resources section.

CountThings for counting total pollen grains

We validated this method to be accurate with a paired t test comparing automated CountThings app counts with hand counts of the same images (24 h time point from one assay). The macro counts and hand counts were not significantly different ($N = 20$, $p = 0.86$). The 24 h time point was selected because the pollen tubes and burst grains present at 24 h (in some images) were can pose an obstacle for accurate automated counting. This script was used for Figure S3 to count total pollen grains. Germinated and burst grains were counted by hand in FIJI as described above.

Protein release and validation in ecologically relevant media

To test whether pollen releases protein into solution when exposed to SCC477, we used a Pierce Modified Lowry Assay Kit (Thermo-Fisher) to quantify protein release into solution from the assay described above in *Confirmation of phenotype in alternate media*. Immediately after mixing of pollen and microbe treatments and at 24 h, 15 μ L was removed with a multichannel pipette and combined with 45 μ L sterile water (for a 1:4 dilution). This was then filtered through a 96 well filter plate (0.2 micron, spun at 1,400 rpm for 4 min) to remove both microbes and pollen. 40 μ L of this solution was then used to quantify protein concentration as per kit instructions; we included a dilution series of known concentration albumin standards in each plate. After reading the plate at 750 nm on a plate reader, the blank (water) value was subtracted from all other values. The albumin dilution series was used to create a standard curve for each plate, which was used to translate 750 nm reading to protein concentration (μ g/mL). Dataset and code are available in the key resources section.

Microbial identification via MALDI-TOF

Because effective sterilization of the pollen could not be done without compromising germinability, we wanted to confirm that added microbes remained as the primary species/strain in their wells, and to identify pollen-borne microbes in the control wells. We therefore validated microbial identity with MALDI-TOF for some of the assays⁸⁶

Within 1 day after select assays (2x microbial impact on pollen assays, 1x dose dependence assay), used wells (50 μ L) were plated onto separate plates of TSA and allowed to grow for 48–72 h. For each morphotype present, (if more than one) material from one or two representative colonies was selected and spotted in duplicate onto an MTP 384 Ground Steel Target Plate (Bruker Daltonics), overlaid with 1 μ L 70% (v/v) formic acid, and allowed to air dry. 1 μ L of alpha-cyano-4-hydroxycinnamic acid (HCCA) matrix dissolved in a 50:45:5 (v/v) solution of acetonitrile, water, and trifluoroacetic acid was added and allowed to dry [same prep as in⁸⁷]. Spectra were obtained twice per spot, in succession, using an ultrafleXtreme MALDI-TOF instrument (Bruker Daltonics, Billerica, MA, United States). Spectra for each isolate were then compared with a custom in-house library of main spectral profiles (MSPs) and the Bruker libraries (Bacteria, Eukaryotes) using MBT Compass Explorer software (Bruker Daltonics, Billerica, MA, United States). Log scores between 1 and 3 were generated for each isolate’s top 10 best matches. Scores ≥ 2.0 are considered close matches; lower scores indicate that the exact species may not have been present or matched closely enough to the library spectra. Genus-level identification was used for those with scores above 1.6. Sometimes, there is no spectra generated from a spot (e.g., HCCA crystals do not form correctly, the spotting is too thick, the laser strikes areas where either of these causes lack of ionization). Spots/wells that failed to generate spectra were removed from the compiled ‘MALDI_identifications.csv’.

Pectinase assay

Microbial strains described above were grown on either Reasoner’s 2 Agar (R2A) or YM plates. For pectin degradation experiments, microbial strains were spotted onto polygalacturonic acid (PGA) plates containing 0.5% PGA in 0.1 M Tris-HCl pH 8.6, with 1.5% agar and 0.5% yeast extract- all (w/v). Plates were incubated for 48–72 h at 25°C and stained with 1% (w/v) cetyltrimethylammonium bromide (CTAB) for 20 min to visualize clearance zones. Each microbe spot was replicated 9 times (Table S1).

To determine whether exposure to pollen would increase or trigger polygalacturonase activity, we grew *A. pollinis* SCC477, *Pectobacterium carotovorum* (positive control) and *Metchnikowia reukaufii* (negative control) on PGA plates. We then mixed one colony of each with prepared fresh *E. californica* pollen in BK germination media, in PCR tubes, to provide initial exposure. As a control for polygalacturonase activity from pollen germination itself, we included another tube into which we added a crushed stigma of *E. californica*, as well as a tube with only pollen in BK media. After one h of co-incubation at room temperature, we checked each treatment for germination and spotted each of these treatments onto fresh PGA plates in triplicate. We also spotted, in triplicate, the microbes themselves directly from the PGA plates that they had been grown on. After incubation for 72 h at 25°C, CTAB solution (1% w/v) was added for 20 m to visualize clearance zones (Figure S2, Table S1).

Examination of flower sterility and identification of common contaminants

We verified that flowers collected prior to opening that opened naturally in lab were relatively free of contaminants. We plated the anthers by removing them with sterilized tweezers, rubbing across surface of plate to spread microbes and pollen, and leaving

them on the plate. We did this for both lab-opened and outside-opened flowers onto multiple media types (3 types: R2A, TSA, and YM). We then picked and identified morphotypes as described above (Figure S4).

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses were done in RStudio 1.2.1335⁷⁹ using R 3.6.2.⁸⁰ Details for each test can be found in figure legends, including the statistical tests used, exact value and definition of *n*, definitions of boxplot parameters (e.g., median, IQR). Significance markings (stars) are defined in Figure 2, significance lettering defined and used in Figures 3 and 4, with baseline threshold for significance $p < 0.05$. Further information can be obtained by running the code for each figure.

To compare if pollen germination and bursting differed among microbes, Kruskal-Wallis tests were run (base R) with microbe treatment as predictor of pollen germination or bursting, respectively. Separate tests were performed for each time point. To test for differences in pollen germination (and separately, pollen bursting) between microbes, pairwise comparisons between microbes were calculated using the “FSA” package⁸⁸ to run a Dunn test with Benjamini-Hochberg *p* value adjustment (28 tests). Eleven total assays were done, and two were eliminated from further analysis. One assay was eliminated due to data loss of images-making recount and validation of pollen ratios impossible, and the other was eliminated due to contamination of the initial growth media. For germination and bursting comparison across dilutions of *A. pollinis* SCC477 (density-dependence), one-way ANOVA was calculated at each time point and the “emmeans” package⁸⁹ was used for pairwise comparisons and calculation of *p* values, with Tukey adjustment (21 tests). For comparison of 24 h cell counts of both *A. pollinis* and *M. reukaufii* (separately) with different pollen treatments, one-way ANOVA was calculated, then the “emmeans” package⁸⁹ was used for pairwise comparisons and calculation of *p* values, with Tukey adjustment (10 tests). To compare protein release over pollen and microbe treatments, a Kruskal-Wallis test was run (base R), with treatment (combined pollen treatment and microbe treatment) as predictor of protein in solution. Pairwise comparisons were calculated using the “FSA” package⁸⁸ using a Dunn test with Benjamini-Hochberg *p* value adjustment (15 tests). To compare germination and (separately) bursting of pollen in artificial nectar, a Kruskal-Wallis test was run (base R), with treatment (combined pollen treatment and microbe treatment) as predictor of germination or bursting. Pairwise comparisons were calculated using the “FSA” package⁸⁸ using a Dunn test with Benjamini-Hochberg *p* value adjustment (6 tests).